

TITLE: IMPLEMENTATION OF ELLIPSE GENERATION ALGORITHM USING MIDPOINT METHOD IN C.

OBJECTIVE:

- To implement the Midpoint Ellipse Generation Algorithm in C and generate the pixel activation list for drawing an ellipse with given center (xc,yc) and radii rx and ry.

THEORY:

An ellipse is a closed curve defined by the equation:

$$(x^2/r_x^2) + (y^2/r_y^2) = 1$$

where:

- r_x = radius along x-axis
- r_y = radius along y-axis

The Midpoint Ellipse Algorithm is an efficient method for rasterizing an ellipse using incremental calculations. It is based on the idea of determining whether the midpoint between two candidate pixels lies inside or outside the ellipse. The algorithm uses decision parameters to select the next pixel position.

ALGORITHM:

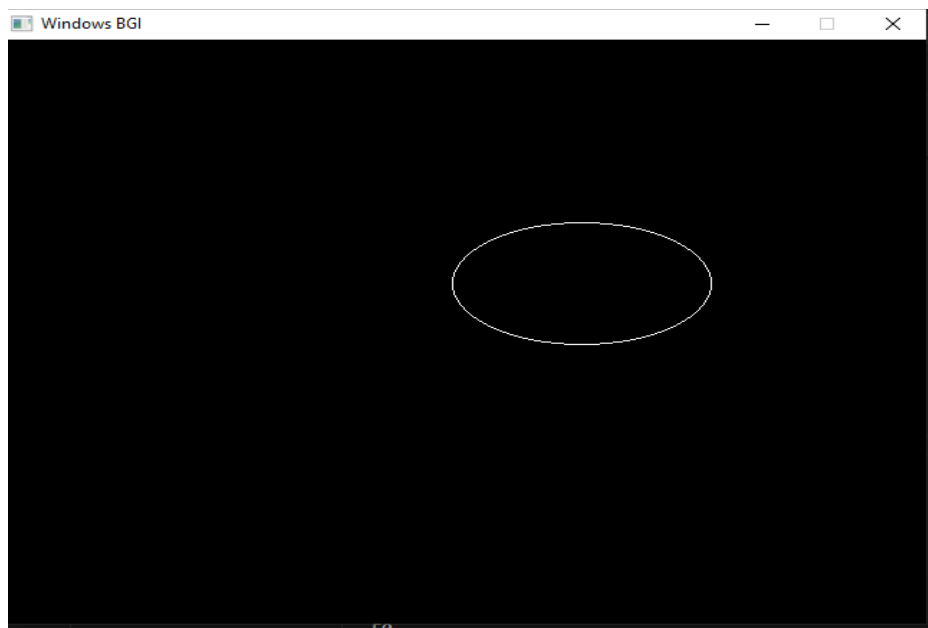
1. Input rx, ry and ellipse center (xc, yc). Obtain the first point on the ellipse centered on the origin as:
 $(x_0, y_0) = (0, r_y)$
2. Calculate the initial decision parameter for Region 1 as:
 $p_1 = (r_y * r_y) - (r_x * r_x * r_y) + (1/4 * r_x * r_x)$
3. At each x position, starting at x = 0:
If $p_1 < 0$, the next point is (x + 1, y) and
 $p_1 = p_1 + (2 * r_y * r_y * x) + (r_y * r_y)$
Otherwise, the next point is (x + 1, y - 1) and
 $p_1 = p_1 + (2 * r_y * r_y * x) - (2 * r_x * r_x * y) + (r_y * r_y)$
Continue this process until:
 $(2 * r_y * r_y * x) \geq (2 * r_x * r_x * y)$
4. The last point calculated above is the end of Region 1.
Calculate the initial decision parameter for Region 2 as:
 $p_2 = (r_y * r_y * (x + 0.5) * (x + 0.5)) + (r_x * r_x * (y - 1) * (y - 1)) - (r_x * r_x * r_y * r_y)$
5. At each y position:
If $p_2 > 0$, the next point is (x, y - 1) and
 $p_2 = p_2 - (2 * r_x * r_x * y) + (r_x * r_x)$
Otherwise, the next point is (x + 1, y - 1) and

$$p2 = p2 + (2 * ry * ry * x) - (2 * rx * rx * y) + (rx * rx)$$

Continue this process until $y = 0$.

6. For both Region 1 and Region 2, determine the symmetric points in the other three quadrants.
7. Move each calculated pixel position (x, y) to the ellipse center (xc, yc) using:
 $x = x + xc$
 $y = y + yc$
8. Plot the points.

OUTPUT:



COCLUSION:

In this experiment, the Midpoint Ellipse Algorithm was successfully implemented to generate an ellipse of given radii and center coordinates. The program accurately calculated pixel positions in one quadrant using decision parameters and then reflected them to the other quadrants using symmetry.